17 Electric Charges and Field

Charge

1. Smallest unit of charge is
   (*a*) Frankline (*b*) Faraday
   (*c*) Coulomb (*d*) e.m.u
2. Which of the following is not the unit of charge?
   (*a*) Farad (*b*) Coulomb
   (*c*) Stat coulomb (*d*) Faraday
3. Neutral body can be negatively charged by?
   (*a*) Giving some excess proton
   (*b*) Giving some extra neutron
   (*c*) Giving some extra electron
   (*d*) Removing some proton
   (*e*) Removing some electron
4. A positively charged body can be made neutral by
   (*a*) Giving some electron (*b*) Giving some proton
   (*c*) Removing some proton (*d*) None of these
5. During charging of body mass of the body increases then body becomes
   (*a*) Positively charge (*b*) Remains neutral
   (*c*) Negatively charge (*d*) None of these
6. Body A is attracted by the charge +Q then body 'A'
   (*a*) Negatively charged
   (*b*) Positively charged
   (*c*) Neutral
   (*d*) May be negatively charged or neutral
7. Body A is repelled by the charge +Q then body 'A' is
   (*a*) –ve charged (*b*) +ve charged
   (*c*) Neutral (*d*) May be –ve or neutral
8. Sure check for presence of electric charge is
   (*a*) Process of induction
   (*b*) Repulsion between bodies
   (*c*) Attraction between bodies
   (*d*) Frictional force between bodies
9. Electric charges *A* and *B* are attracted to each other. Electric charges *B* and *C* are also attracted to each other. If *A* and *C* are held close together they will
   (*a*) attract
   (*b*) repel
   (*c*) not affect each
   (*d*) More information is needed to answer
10. Two charge θ_1 and θ_2 shows repulsion then which of the following option is correct for this:-
    (*a*) $\theta_1 \times \theta_2 = 0$ (*b*) $\theta_1 \times \theta_2 > 0$
    (*c*) $\theta_1\theta_2 < 0$ (*d*) all of these
11. Compare specific charge of neutron, proton, α-particle

Quantisation of Charge

12. **Assertion (A):** Charge is quantized
    Reason (R): Charge less than 1C is not possible
    (*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
    (*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A)
    (*c*) Assertion (A) is true but Reason (R) is false
    (*d*) Assertion (A) is false and Reason (R) is true
13. Which of the following charge is possible on object
    (*a*) 24×10^{-18} C (*b*) 0.1 C
    (*c*) 0.1 e (*d*) 32×10^{-20}C
14. Which of the following charge is possible :
    (*a*) $\frac{1C}{100}$ (*b*) $\frac{1e}{50}$
    (*c*) 4.8×10^{-21} C (*d*) 1.56×10^{-15} C
15. Which of the following charge is not possible?
    (*a*) 14.4×10^{-19} (*b*) 64×10^{-20}
    (*c*) 3.2×10^{-20} (*d*) 1678×10^{-14} C
16. Charge developed on the body if 10 e^- is removed from that body

17. Charge on α-particle is
(a) 4.8×10^{-19} C (b) 1.6×10^{-19} C
(c) 3.2×10^{-19} C (d) 6.4×10^{-19} C

18. Find no of electron removed from body to charge it by + 1C.

19. When 10^{14} electrons are removed from a neutral metal sphere, then the charge on the sphere becomes
(a) 16 μC (b) –16 μC (c) 32 μC (d) –32 μC

20. A glass rod is rubbed with a silk cloth. The glass rod acquires a charge of 19.2×10^{-19}C. Find the number of electrons lost by glass rod.

21. A conductor has 14.4×10^{-19}C positive charge. The conductor has (charge on electron = 1.6×10^{-19} C)
(a) 9 electrons in excess (b) 27 electrons in short
(c) 27 electrons in excess (d) 9 electrons in short

22. Charge on a body is –1 PC then find excess e^- given to that object. Also find mass of extra electron given to that charge during charging.

23. If 10^{10} electron are acquired by a body per sec then find time in which body will get total charge 1C ?

24. Find total charge on 5kg electrons.

25. Which of the following is correct regarding electric charge?
(i) If a body is having positive charge, i.e., shortage of electrons
(ii) If a body is having negative charge i.e., excess of electrons
(iii) Minimum possible charge = ± 1.6×10^{-19} C
(iv) Charge is quantized, i.e., Q = ± ne, where n = 1, 2, 3, 4, ...
(a) Both (i) and (ii) (b) Both (ii) and (iii)
(c) (i), (ii), (iii) (d) All of these

26. Speed of a charged body is decreasing continuously then its charge and specific charge will
(a) Increase decrease (b) same, increase
(c) same, decrease (d) same, same

Properties of Electric Charge

27. Which of the following is not true about electric charge?
(a) Charge is a scalar quantity
(b) Charge on an isolated system is always conserved
(c) A particle having nonzero rest mass can have zero charge
(d) A particle having zero rest mass can have non-zero charge

28. Select the correct statement about electric charge
(a) Charge can be converted into energy and energy can be converted into charge
(b) Charge of a particle increases with increase in its velocity
(c) Charge on a body Is always integral multiple of a certain charge called charge of electron
(d) Charge on a a body is always positive or zero

29. Identify incorrect statements for electric charge *q*:-
(a) Quantised (b) Conserved
(c) Additive (d) Non-transferable

Conductors and Insulators

30. If a solid and a hollow conducting sphere have same radius then
(a) Hollow sphere will hold more maximum charge
(b) Solid sphere will hold more maximum charge
(c) Both the spheres will hold same maximum charge
(d) Both the sphere can't hold charge

31. If an object made of substance A rubs an object made of substance B, then A becomes positively charged and *B* becomes negatively charged, If. however, an object made of substance A is rubbed against an object made of substance C. then A becomes negatively charged. What will happen if an object made of substance B is rubbed against an object made of substance C?
(a) B becomes positively charged and C becomes positively charged.
(b) B becomes positively charged and C becomes negatively charged.
(c) B becomes negatively charged and C becomes positively charged.
(d) B becomes negatively charged and C becomes negatively charged.

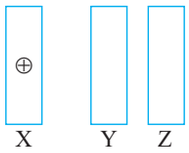
32. Two metallic spheres of radii 1 cm and 3 cm are given charges of -1×10^{-2}C and 5×10^{-2}C, respectively. If these are connected by conducting wire, the final charge on the bigger sphere is :
(a) 4×10^{-2} C (b) 1×10^{-2} C
(c) 2×10^{-2} C (d) 3×10^{-2} C

33. Two sphere of Radius R and 4R having charge 15C and –5C respectively, if they are placed in contact then find final charge on each sphere.

34. When three identical sphere X, Y and Z having charge Q, –2Q and 3Q respectively. At first X and Y are in contact then Y and Z are in contact then final charge on X, Y and Z will be
(a) $\frac{Q}{2}, \frac{3Q}{2}, \frac{Q}{2}$ (b) $\frac{-Q}{2}, \frac{5Q}{4}, \frac{+5Q}{4}$
(c) Q, Q, Q (d) $\frac{Q}{2}, \frac{3Q}{2}, \frac{-Q}{2}$

35. If point charge is placed very near to the uncharged conducting sphere then net charge induced on sphere is
(a) Zero (b) +ve
(c) –ve (d) can't say

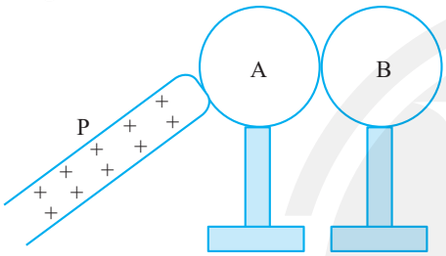
36. Identify nature of force between body X and Y and Y and Z (Y and Z are neutral)



(*a*) Attract, repel
(*b*) repel, attract
(*c*) repel, repel
(*d*) attract, attract

37. Electric charges A and B repel each other. Electric charges B and C also repel each other. If A and C are held close together they will
(*a*) attract
(*b*) repel
(*c*) not affect each other
(*d*) More information is needed to answer

38. Two uncharged metal sphere placed in contact as shown in figure and A positive charged rod placed in contact with A, then charge on A and B will be



(*a*) Both become +ve
(*b*) Both become -ve
(*c*) A is +ve but B is –ve
(*d*) DA is –ve but B is +ve

39. **Assertion (A):** When bodies are charged through friction, there is transfer of charge from one body to another. No charge is created or destroyed.

Reason (R): This is according to the law of conservation of electric charge

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A)
(*c*) Assertion (A) is true but Reason (R) is false
(*d*) Assertion (A) is false and Reason (R) is true

40. **Assertion (A):** Charging is due to transfer of electrons

Reason (R): Mass of a body decreases slightly when it is negatively charged

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
(*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A)
(*c*) Assertion (A) is true but Reason (R) is false
(*d*) Assertion (A) is false and Reason (R) is true

Gold Leaf Electroscope

41. What is the application of a gold leaf electroscope?
(*a*) To measure temperature
(*b*) To detect charge and identify its nature
(*c*) To determine pressure
(*d*) To identify the mass of an object

42. When *a* + 20 C rod is touched to the knob, the deflection of leaves was 5°, and when an identical rod of –40 C is touched, the deflection was found to be 9°. If an identical rod of +30 C is touched, then the deflection may be :
(*a*) 0 (*b*) 2°
(*c*) 7° (*d*) 11°

43. A glass rod rubbed with silk is touched with the disc of a negatively charged gold leaf electroscope. The divergence of leaves will
(*a*) decreases
(*b*) increases
(*c*) remain uncharged
(*d*) first decreases and then increase

44. A gold leaf electroscope is given a positive charge so that its leaves diverge. How is the divergence of leaves affected, when a glass rod rubbed with silk is rolled on the disc of electroscope?
(*a*) divergence increases
(*b*) divergence decreases
(*c*) divergence remains same
(*d*) can't say

45. A charged gold leaf electroscope has its leaves apart by a certain amount of having enclosed air. When the electroscope is subjected to X-rays, then the leaves:
(*a*) Further dilate (*b*) Start oscillating
(*c*) Collapse (*d*) Remain unaltered

Coulomb's Law : Electrostatic Force

46. Find minimum electrostatic force between two electrons (1.6×10^{-19}C) places at 1m distance?

47. Two proton placed at *r* distance then find ratio of electrostatic to gravitational force acting between them?

48. Two point charge separated by a distance r repel each other with a force of 64 N, if distance between them becomes 4-times and each charges becomes double then new force between them will be.

49. Two charge of 1 nC are at distance 3 cm apart in vacuum, then the force between them is

50. Two sphere of radius in the radio 1 : 2 having equal and opposite charges exert a force F, now they placed in contact and again placed at previous position then new force acting on them will be?
(*a*) F (*b*) F/2 (*c*) 2F (*d*) Zero

51. The charge on two sphere is 8C and –4C respectively they experience a force F, if each charge is given –4C on each sphere then new force between them is

(*a*) F (*b*) F/2 (*c*) F/4 (*d*) 2F

52. Two point charges repel each other with a force of 100 N. One of the charges is increased by 10% and other is reduced by 10%. Then new force of repulsion at the same distance would be:

(*a*) 99 N (*b*) 121 N (*c*) 100 N (*d*) 110 N

53. Two point charges A and B, having charges +Q and -Q respectively, are placed at certain distance apart and force acting between them is F. If 25% charge of A is transferred to B, then force between the charges becomes

(*a*) $\frac{4F}{3}$ (*b*) F (*c*) $\frac{9F}{16}$ (*d*) $\frac{16F}{9}$

54. Two small conducting sphere having charge 8C and –2C respectively and placed at distance *R*, then force between them is F_1, if they are brought in contact and separated to the same distance then force F_2 then ratio of F_1: F_2 is

55. Two charges +2C and +6 C are repelling each other with a force of 12 N. If each charge is given –2 C of charge, then the value of the force will be:

(*a*) 4 N (attractive) (*b*) 4 N (repulsive)

(*c*) 8 N (repulsive) (*d*) zero

56. A charge q is to be divided on two small conducting spheres. What should be the value of charges on the spheres so that when placed at a certain distance apart, the repulsive force between them is maximum?

(*a*) $\frac{q}{4}$ and $\frac{3q}{4}$ (*b*) $\frac{q}{2}$ and $\frac{q}{2}$ (*c*) $\frac{q}{3}$ and $\frac{q}{3}$ (*d*) $\frac{q}{4}$ and $\frac{q}{4}$

57. Two positive ions, each carrying a charge q, are separated by a distance *d*. If F is the force of repulsion between the ions, the number of electrons missing from each ion will be (*e* being the charge on an electron)

(*a*) $\frac{4\pi\varepsilon_0 Fd^2}{e^2}$ (*b*) $\sqrt{\frac{4\pi\varepsilon_0 Fe^2}{d^2}}$

(*c*) $\sqrt{\frac{4\pi\varepsilon_0 Fd^2}{e^2}}$ (*d*) $\frac{4\pi\varepsilon_0 Fe^2}{q^2}$

58. A particle of mass m and carrying charge $-q_1$ is moving around a charge $+q_2$ along a circular path of radius *r*. Find period of revolution of the charge $-q_1$

(*a*) $\sqrt{\frac{16\pi^3\varepsilon_0 mr^3}{q_1q_2}}$ (*b*) $\sqrt{\frac{8\pi^3\varepsilon_0 mr^3}{q_1q_2}}$

(*c*) $-\sqrt{\frac{q_1q_2}{16\pi^3\varepsilon_0 mr^3}}$ (*d*) Zero

59. Two point object having equal mass & equal and opposite charge q moving on circular path due to their mutual electrostatic force then find their speed ?

60. Two identical spheres are given equal charges in one case & opposite charge in another case having force F_1 & F_2 respectively. Distance between spheres is not large as compare to diameter then compare F_1 & F_2.

(*a*) $F_1 > F_2$

(*b*) $F_1 = F_2$

(*c*) $F_1 < F_2$

(*d*) Information is not sufficient to draw the conclusion

61. Suppose the charge of a proton and an electron differ slightly. One of them is –e, the other is ($e + \Delta e$). If the net of electrostatic force and gravitational force between two hydrogen atoms placed at a distance *d* (much greater than atomic size) apart is zero, then Δe is of the order of [Given mass of hydrogen m_h = underline 1.67×10^{-27} kg]

(*a*) 10^{-23} C (*b*) 10^{-37} C (*c*) 10^{-47} C (*d*) 10^{-20} C

62. A rod of length *L* and linear charge density λ is placed along *x*-axis and a point charge *q* is placed and distance '*a*' from one end then find force on rod due to this point charge.

63. Force between two point charges q_1 and q_2 placed in vacuum at '*r*' cm apart is *F*. Force between them when placed in a medium having dielectric constant K = 5 at '*r*/5' cm apart will be:

(*a*) 5F (*b*) 25F (*c*) F/5 (*d*) F/25

64. Two identical charged particles each having a mass 10*g* and charge 2×10^{-7} C are placed on a horizontal table with separation of *L* between them such that they stay in limited equilibrium. If coefficient of friction between each particle and table is 0.25, find value of *L*

(*a*) 12 cm (*b*) 10 cm (*c*) 8 cm (*d*) 5 cm

65. **Assertion (A):** Coloumb's law is valid only for point charges and spherical symmetry.

Reason (R): Electrostatic force is conservative in nature

(*a*) Assertion (A) is true but Reason (R) is false

(*b*) Assertion (A) is false but Reason (R) is true

(*c*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A)

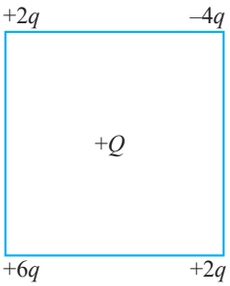
(*d*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

66. A particle having charge q_1 exert force *F* on charge q_2, If a particle having charge $-2q_1$ is placed midway between the line then new force will be on q_2

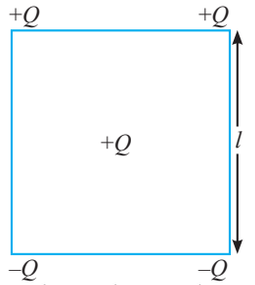
(*a*) 4F (*b*) 2F (*c*) 6F (*d*) 7F

67. Four identical charge *Q* placed at the corner of square then find net force on 5th charge at centre due to all four charges

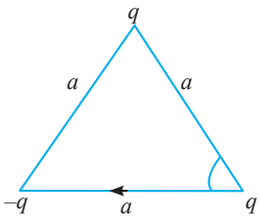
68. Find net force on +*Q* charge which is at centre of square of length *l*.



69. Find net force on *Q* charge which is at centre of square of length *l*.



70. Find force on +q charge due to other two charge

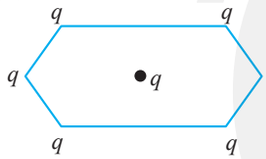


(*a*) $\frac{\sqrt{3}Kq^2}{a^2}$ (*b*) $\frac{Kq^2}{a^2}$

(*c*) Zero (*d*) $\frac{2Kq^2}{a^2}$

71. Four point charge +Q placed on the corner of square of length l then find net electrostatic force on any one charge due to remaining three charges

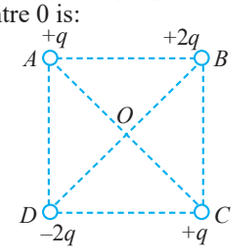
72. Find net force on charge q which is at centre of hexogonal of side length l.



(*a*) $\frac{5Kq^2}{l^2}$ (*b*) $\frac{Kq^2}{l^2}$

(*c*) $\frac{2Kq^2}{l}$ (*d*) Zero

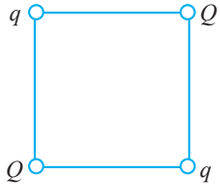
73. Four charges are arranged at the corners of a square *ABCD*, as shown in the adjoining figure. The force on the charge kept at the centre 0 is:



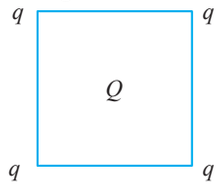
(*a*) Zero (*b*) Along the diagonal *AC*

(*c*) Along the diagonal *BD* (*d*) Perpendicular to side *AB*

74. Find relation between 'q' and 'Q' so that force on Q is zero.



75. Find value of Q so that force on charge q is zero which is at corner of square of figure.



76. Find force on q charge which is at origin due to all other charges

q_0	q	q	q	q	upto infinity
(0,0)	(1,0)	(2,0)	(4,0)	(8,0)	(16,0) soon

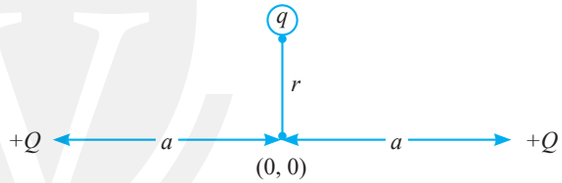
77. Find force on q charge which is at origin due to all other charges

q_0	$+q$	$-q$	$+q$	$-q$	upto infinity
(0,0)	(1,0)	(2,0)	(4,0)	(8,0)	(16,0) soon

78. Two equally charged, identical metal spheres A and B repel each other with a force 'F'. The spheres are kept fixed with a distance 'r' between them. A third identical, but uncharged sphere C is brought in contact with A and then placed at the mid-point of the line joining A and B. The magnitude of the net electric force on C is

(*a*) F (*b*) $3F/4$ (*c*) $F/2$ (*d*) $F/4$

79. Find force on 'q' due to two +Q charge as shown in figure.



80. A neutral bubble made of soap solution has radius R. If some negative charge is given which gets distributed uniformly on the surface of the bubble, the radius of the bubble becomes R' then

(*a*) $R' > R$ (*b*) $R' = R$ (*c*) $R' < R$ (*d*) $R' = 0.5\,R$

Coulomb's Law in Vector Form

81. Find force applied by 2C charge which is at (2, 3, 4) on 5C charge which is at (5, 7, 9)

82. Find force on –3C charge at (2, 0, 4) due to 4C charge which is at (5, 4, 4).

83. An electron is moving round the nucleus of a hydrogen atom in a circular orbit of radius r. The Coulomb force $\vec{F}$ on the electron is:

(*a*) $K\frac{e^2}{r^2}\hat{r}$ (*b*) $-K\frac{e^2}{r^2}\hat{r}$ (*c*) $K\frac{e^2}{r^3}\hat{r}$ (*d*) $-K\frac{e^2}{r^3}\hat{r}$

Superposition Theorem

84. Electrostatic force on charge does not depends on medium.

(*a*) True (*b*) False

85. Electric force on one charge due to other charge does not depends on medium or presence of other charge

(*a*) True (*b*) False

86. **Statement I:** Electrostatic force on q_2 due to q_1 doesn't depend upon medium.

Statement II: Net electrostatic force on q_2 depends upon medium but not on presence of other charges.

(*a*) I is true but II is false
(*b*) I is false but II is true
(*c*) Both I and II are true but II is not the correct explanation of I
(*d*) Both I and II are true and II is the correct explanation of I

87. Find position of 3rd charge q so that force on q will be zero.

(*i*) Q ⟵ 10m ⟶ $+16Q$
(*ii*) $+3Q$ ⟵ 30m ⟶ $27Q$
(*iii*) $25Q$ ⟵ 20m ⟶ $16Q$
(*iv*) $-4Q$ ⟵ 10m ⟶ $-8Q$

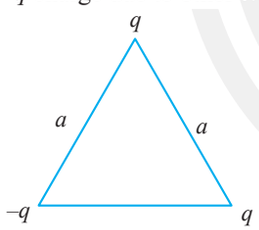
88. Find position of 3rd charge so that force on 3rd charge will be zero.

$25Q$ ⟵ 6m ⟶ $-9Q$

89. Find position of 3rd charge so that force on 3rd charge will be zero.?

(*i*) $-Q$ ⟵ $r = 10$ m ⟶ $-4Q$
(*ii*) $-2Q$ ⟵ $r = 10$ m ⟶ $+8Q$

90. Find force on $+q$ charge due to other two charge.



(*a*) $\frac{\sqrt{3}Kq^2}{a^2}$ (*b*) $\frac{Kq^2}{a^2}$ (*c*) Zero (*d*) $\frac{2Kq^2}{a^2}$

91. Nature of equilibrium of $-q$ charge placed at mid of two $+Q$ charge

$+Q$ ---------- $-q$ ---------- $+Q$

92. Nature of equilibrium of $-q$ charge placed at mid of two $+Q$ charge

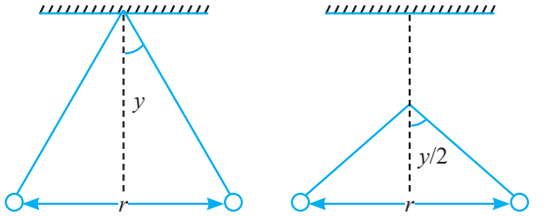
$+Q$ ---------- $-q$ ---------- $+Q$

93. Consider two identical charges placed at distance $2d$ apart. The $-q$ charge placed at centre of line joining two charges. Which of the following is not incorrect statement?

(*a*) The $-q$ charge is not in equilibrium
(*b*) $-q$ charge will oscillate if it is displaced slightly along x-axis
(*c*) $-q$ charge will oscillate if it is displaced slightly along y-axis
(*d*) The $-q$ charge is in stable equilibrium along x-axis

Pendulum Problem

94. Two pith balls carrying equal charges are suspended from a common point by strings of equal length, the equilibrium separation between them is r. Now the strings are rigidly clamped at half the height. The equilibrium separation between the balls now become

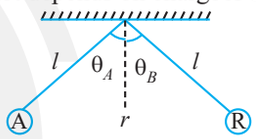


(*a*) $\left(\frac{2r}{3}\right)$ (*b*) $\left(\frac{1}{\sqrt{2}}\right)^2$ (*c*) $\left(\frac{r}{\sqrt[3]{2}}\right)$ (*d*) $\left(\frac{2r}{\sqrt{3}}\right)$

Electric Field

95. **Assertion (A):** In the given figure, $\theta_A = \theta_B$ even if $Q_A > Q_B$ (provided $m_A = m_B$)

Reason (R): Angle of deflection of one individual particle A or B does not depends on charge A & B



(*a*) Assertion (A) is true but Reason (R) is false
(*b*) Assertion (A) is false but Reason (R) is true
(*c*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A)
(*d*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

96. The magnitude of electric field strength E such that an electron placed in it would experience an electric force equal to its weight is given by

(*a*) mge (*b*) $\frac{mg}{e}$ (*c*) $\frac{e}{mg}$ (*d*) $\frac{e^2}{mg}$

97. The magnitude of force exerted by a uniform electric field on an electron having mass m_e and proton of mass m_p are represented as F_e and F_p respectively are related as

(*a*) $F_p = F_e$ (*b*) $\frac{F_e}{F_p} = \frac{m_e}{m_p}$ (*c*) $\frac{F_e}{F_p} = \frac{m_p}{m_e}$ (*d*) $\frac{F_e}{F_p} = \frac{m_p}{m_e}$

98. **Statement I:** Force experienced by unit charge (+ or –) is called electric field intensity.

Statement II: Electric field can form closed loop.

(*a*) Both correct
(*b*) Both incorrect
(*c*) Both I and II are true but II is not the correct explanation of A
(*d*) Both I and II are true and II is the correct explanation of A

99. The electric field at a point is:

(a) Always continuous X

(b) Continuous if there is no charge at that point

(c) Discontinuous only if there is a negative charge at the pointX

(d) Discontinuous if there is a charge at that pointy

(*a*) (a, b) (*b*) (b, d) (*c*) (c, d) (*d*) (a, d)

100. The electrostatic field due to a charged conductor just outside the conductor is:

(*a*) zero and parallel to the surface at every point inside the conductor

(*b*) zero and is normal to the surface at every point inside the conductor

(*c*) parallel to the surface at every point and zero inside the conductor

(*d*) normal to the surface at every point and zero inside the conductor

101. **Assertion (A):** In a cavity in a conductor, the electric field is zero

Reason (R): Charges in a conductor reside only at its surface

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

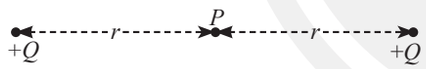
(*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A)

(*c*) Assertion (A) is true but Reason (R) is false

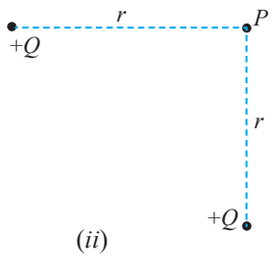
(*d*) Assertion (A) is false and Reason (R) is true

Combination of Point Charges

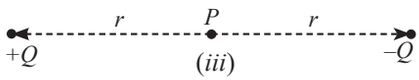
102. Electric field due to combination of point charge at Point *P*



(*i*)

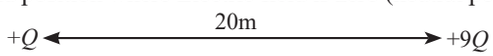


(*ii*)



(*iii*)

103. Find position where Electric field is zero (neutral point)



104. Charge 2Q and –*Q* are placed as shown in figure. The point at which electric field intensity is zero will be somewhere



(*a*) Between –*Q* and 2*Q*

(*b*) On the left of –*Q*

(*c*) On the right of 2*Q*

(*d*) On the perpendicular bisector of line joining the charges

105. A point charge of 10 μ*C* is placed at the origin. At what location on the *x*-axis should a point charge of 40 μ*C* be placed so that the net electric field is zero at *x* = 2 cm on the *x*-axis?

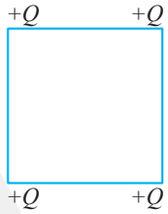
(*a*) *x* = 6 cm (*b*) *x* = 8 cm (*c*) *x* = 4 cm (*d*) *x* = –4 cm

106. The distance between the two charges 25 μ*C* and 36 μ*C* is 11 cm. At what point on the line joining the two charges the intensity will be zero at a distance of

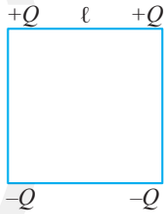
(*a*) 4 cm from 36 μC (*b*) 4 cm from 25 μC

(*c*) 5 cm from 36 μC (*d*) 5 cm from 25 μC

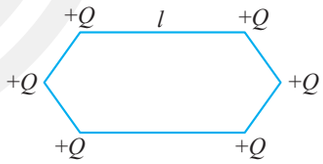
107. Find the electric field at the centre of square.



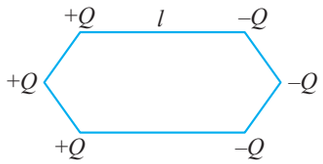
108. Electric field at centre?



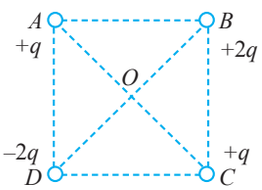
109. Find electric field at the centre.



110. Find electric field at the centre.

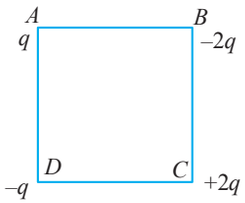


111. Four charges are arranged at the corners of a square *ABCD*, as shown in the adjoining figure. The force on the charge kept at the centre *O* is:

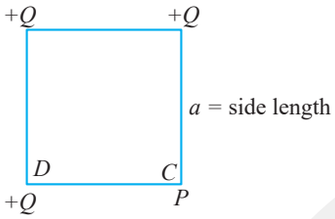


(a) Zero
(b) Along the diagonal AC
(c) Along diagonal BD
(d) Perpendicular to side AB

112. Four charges are placed on corners of a square as shown in figure having side 10 cm. If q is 1 μC then what will be electric field intensity at the centre of the square?



113. Find electric field at 'P'



114. A charge Q is placed at the centre of a square. If electric field intensity due to the charge at the corners of the square is E_1 and the intensity at the mid point of the side of square is E_2, then the ratio of E_1/E_2 will be

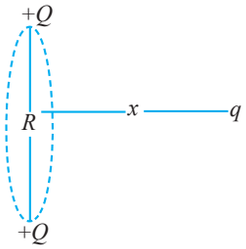
(a) $\frac{1}{2\sqrt{2}}$ (b) $\sqrt{2}$ (c) $\frac{1}{2}$ (d) 2

115. 7 Point charge placed at the corner of cube find electric field at centre of cube.

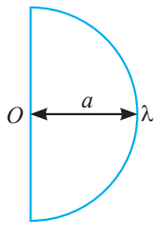
Electric Field Due to Ring

116. Electric field due to ring of radius R on the axis of ring is maximum at a distance of

117. Find value of x so that force on q will be maximum?

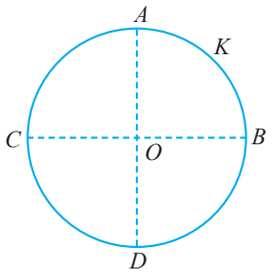


118. Electric field at centre O of semicircle of radius a having linear charge density λ given by



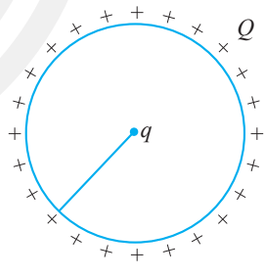
(a) $\frac{2\lambda}{\varepsilon_0 a}$ (b) $\sqrt{2}$
(c) $\frac{\lambda}{2\pi\varepsilon_0 a}$ (d) $\frac{\lambda}{\pi\varepsilon_0 a}$

119. A thin conducting ring of radius R is given a charge $+Q$. The electric field at the centre O of the ring due to the charge on the part AKB of the ring is E. The electric field at the centre due to the charge on the part $ACDB$ of the ring is:



(a) 3E along OK (b) 3E along KO
(c) E along OK (d) E along KO

120. A positive charge Q is uniformly distributed along a circular ring of radius R, A small test charge q is placed at the centre of the ring. Then,
(a) If $q > 0$ and is displaced away from the centre in the plane of the ring, it will be pushed back towards the centre
(b) If $q < 0$ and is displaced away from the centre in the plane of the ring, it will never return to the centre and will continue moving till it hits the ring.
(c) If $q < 0$, it will perform SHM for small displacement along the axis.
(d) q at the centre of the ring is in an unstable equilibrium within the plane of the ring for $q > 0$.



(a) (a, b, c) (b) (a, c, d) (c) (b, c, d) (d) (c, d)

Electric Field Lines

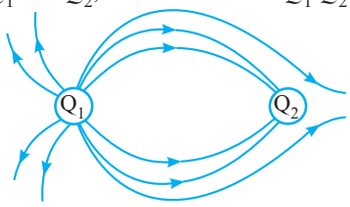
121. Field-lines are always perpendicular at the surface of a conductor but they never enter inside the conductor because there is no electrostatic field inside the material of a conductor.
(a) True (b) False

122. The tangent at any point of a field-line gives the direction of the force on a positive charge at that point.
(a) True (b) False

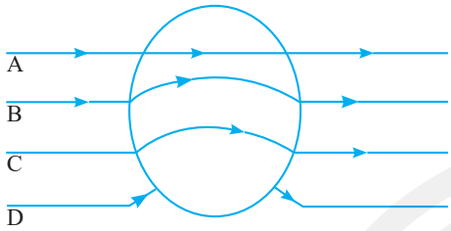
123. Electric field lines always start on positive charges and end on negative charges

(*a*) True (*b*) False

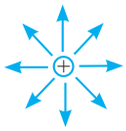
124. Looking at the various electric lines of forces between charges Q_1 and Q_2, find the value of Q_1/Q_2-

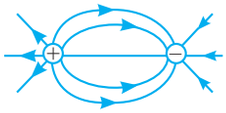


125. A solid metallic sphere is placed in a uniform electric field. Which of the lines *A*, *B*, *C* and *D* shows the correct path and why?



126. Which of the electrostatic lines of forces are correctly drawn ?

(*a*)  (*b*) 

(*c*)  (*d*) All of these

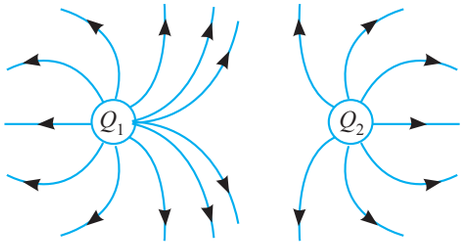
127. The figure shows electric field lines. If E_A and E_B are electric field at *A* and *B* and distance *AB* is *r*, then



(*a*) $E_A > E_B$ (*b*) $E_A = E_B/r$

(*c*) $E_A < E_B$ (*d*) $E_A = E_B/r^2$

128. Figure shows electric lines of forces due to charges Q_1 and Q_2. Hence



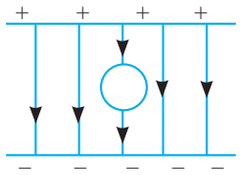
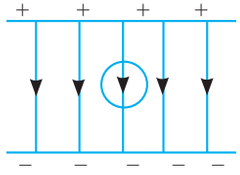
(*a*) Q_1 and Q_2 both are negative

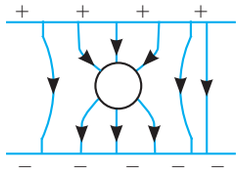
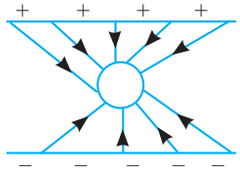
(*b*) Q_1 and Q_2 both are positive

(*c*) $Q_1 > Q_2$

(*d*) Both (*b*) and (*c*)

129. An uncharged sphere of metal is placed in a uniform electric field produced by two oppositely charged plates. The lines of force will appear as

(*a*)  (*b*) 

(*c*)  (*d*) 

130. Electric field lines are also called electrostatic force line.

(*a*) True (*d*) False

131. **Assertion (A):** Electric lines of force cross each other.

Reason (R): The resultant electric field at a point is the superimposition of the electric field at that point.

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A)

(*c*) Assertion (A) is true but Reason (R) is false

(*d*) Assertion (A) is false and Reason (R) is true

132. If the number of electric lines of force emerging out of a closed surface is 1000, then the charge enclosed by the surface is

(*a*) 8.854×10^{-9} C

(*b*) 8.854×10^{-4} C

(*c*) 8.854×10^{-1} C

(*d*) 8.854 C

Motion of Charged Particle in Uniform Electric Field

133. When charge particle moves in the direction of electric field its speed will Increase?

(*a*) True (*b*) False

134. A proton of mass '*m*' charge '*e*' is released from rest in a uniform electric field of strength '*E*'. The time taken by it to travel a distance '*d*' in the field is

(*a*) $\sqrt{\frac{2de}{mE}}$ (*b*) $\sqrt{\frac{2dm}{Ee}}$ (*c*) $\sqrt{\frac{2dE}{me}}$ (*d*) $\sqrt{\frac{2Ee}{dm}}$

135. An electron falls from rest through a vertical distance *h* in a uniform and vertically upward directed electric field *E*.

The direction of electric field is now reversed, keeping its magnitude the same. A proton is allowed to fall from rest in it through the same vertical distance h. The time of fall of the electron, in comparison to the time of fall of the proton is

(*a*) Smaller (*b*) 5 times greater

(*c*) Equal (*d*) 10 times greater

136. If electron and proton released in electric field find ratio of momentum after time '*t*'.

137. Deuteron and alpha particle thrown in opposite direction of electric field with same momentum at same time then find ratio of time when they will stop.

(*a*) 1 : 1 (*b*) 2 : 1 (*c*) 1 : 2 (*d*) 4 : 1

138. If electron and proton are released in electric field then find ratio of K.E. after distance x

139. A particle of mass m and charge q is placed at rest in a uniform electric field E and then released. The kinetic energy attained by the particle after moving a distance y is

(*a*) qEy (*b*) qE^2y (*c*) qEy^2 (*d*) q^2Ey

140. A mass m carrying a charge q is suspended from a string and placed in a uniform horizontal electric field of intensity E. The angle made by the string with the vertical in the equilibrium position is

(*a*) $\theta = \tan^{-1}\frac{mg}{Eq}$ (*b*) $\theta = \tan^{-1}\frac{m}{Eq}$

(*c*) $\theta = \tan^{-1}\frac{Eq}{m}$ (*d*) $\theta = \tan^{-1}\frac{Eq}{mg}$

141. A positively charged pendulum is oscillating in a uniform electric field pointing upwards. Its time period as compared to that when it oscillates without electric field

(*a*) Is less (*b*) Is more

(*c*) Remains unchanged (*d*) Starts fluctuating

142. A particle of mass m and charge q is thrown at a speed u against a uniform electric field E. How much distance will it travel before coming to rest?

(*a*) $mu^2/2qE$ (*b*) $2mu^2/qE$ (*c*) $mqE/2u^2$ (*d*) mu/qE

143. A toy car with charge q moves on a frictionless horizontal plane surface under the influence of a uniform electric field $\vec{E}$. Due to the force $q\vec{E}$, its velocity increases from 0 to 6 m/s in one second duration. At that instant the direction of the field is reversed The car continues to move for two more seconds under the influence of this field. The average velocity and the average speed of the toy car between 0 to 3 seconds are respectively

(*a*) 2 m/s, 4 m/s (*b*) 1m / s, 3 m/s

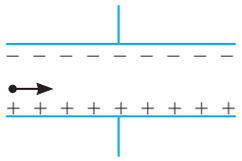
(*c*) 1.5m / s 3 m/s (*d*) 1 m/s, 3.5 m/s

144. In a uniform electric field if a charge is fired in a direction different from the line of electric field then the trajectory of the charge will be a

(*a*) Straight line (*b*) Circle

(*c*) Parabola (*d*) Ellipse

145. A proton and an α particle having equal kinetic energy are projected in a uniform transverse electric field as shown in figure



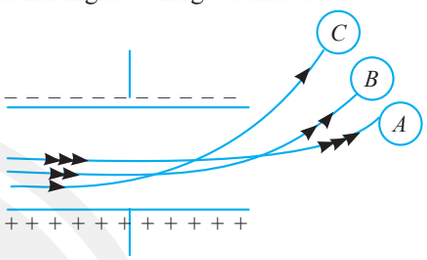
(*a*) Proton trajectory is more curved

(*b*) α particle trajectory is more curved

(*c*) Both trajectories are equally curved but in opposite direction

(*d*) Both trajectories are equally curved and in same direction

146. Three particles are projected in a uniform electric field with same velocity perpendicular to the field as shown. Which particle has highest charge to mass ratio?



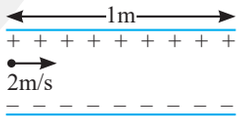
(*a*) A

(*b*) B

(*c*) C

(*d*) All have same charge to mass ratio

147. A uniform electric field $E = (8m / e)$ V/m is created between two parallel plates of length 1 m as shown in figure, (where m = mass of electron and e = charge of electron). An electron enters the field symmetrically between the plates with a speed of 2 m/s. The angle of the deviation (θ) of the path of the electron as it comes out of the field will be_____.



(*a*) $\tan^{-1}(4)$ (*b*) $\tan^{-1}(2)$ (*c*) $\tan^{-1}(1/3)$ (*d*) $\tan^{-1}(3)$

148. Proton, deuteron and α-particle projected in electric field. Perpendicular to it, then find ratio of deviation if:

(*i*) Velocity of projection is same

(*ii*) K.E. of projection same

(*iii*) Projected with same momentum

149. A proton is kept at rest. A positively charged particle is released from rest at a distance S in its field. Consider two experiments; one in which the charged particle is also a proton and in another, a positron. In the same time t, the work done on the two moving charged particles is:

(*a*) Same as the same force law is involved in the two experiments

(*b*) Less for the case of a positron, as the positron moves away more rapidly and the force on it weakens

(*c*) More for the case of a positron, as the positron moves a larger distance, away

(*d*) Same as the work is done by a charged particle on the stationary proton

150. A charge particle is free to move in an electric field. It will travel:

(*a*) Always along a line of force

(*b*) Along a line of force, if its initial velocity is zero

(*c*) Along a line of force, if it has some initial velocity in the direction of an acute angle with the line of force

(*d*) None of the above

151. Electric field line always along the path taken or travelled by the charge placed in electric field.

(*a*) True (*b*) False

152. Two equal positive charges Q are fixed at points $(a, 0)$ and $(-a, 0)$ on the x-axis. An opposite charge -9 at rest is released from point $(0, a)$ on the y-axis. The charge -9 will

(*a*) Move to infinity

(*b*) Move to origin and rest there

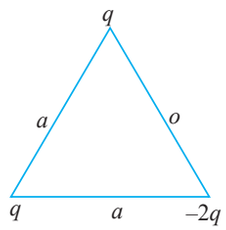
(*c*) Undergo SHM about the origin

(*d*) Execute oscillatory periodic motion but not SHM

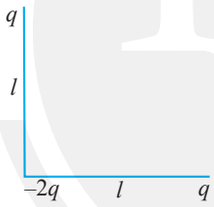
Electric Dipole

153. Find electric dipole moment of system?

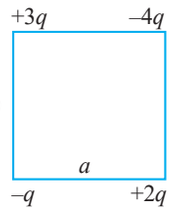
(*i*)



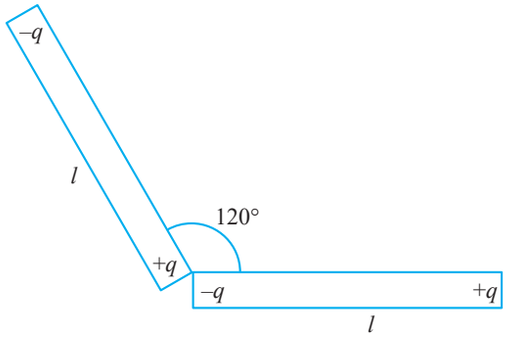
(*ii*)



(*iii*)



154. Find net electric dipole moment?



155. Three point charges q, q and $-2q$ are located at $(0, -1, 1)$, $(0, 1, 1)$ and $(0, 0, -1)$ respectively. The magnitude of electric dipole moment of this charge distribution is :

(*a*) Zero (*b*) q

(*c*) 2q (*d*) 2q

156. Which among the following is the dimensional formula of dipole moment?

(*a*) $[L^1T^1I^1]$ (*b*) $[L^1\,TI^2]$

(*c*) $[M^2\,L^2\,TI^2]$ (*d*) $[L^2\,T^2\,I^{-2}]$

157. The nature of electric flux from the cube surface, if an electric dipole is placed inside a cube will be?

(*a*) The electric flux will be releasing out of the surface

(*b*) The electric flux will be coming into the surface

(*c*) There will be no flux at all

(*d*) None of these

158. 3 C and -3 C are two equal charges separated by a distance of 4 cm, then calculate the dipole moment of a dipole.

(*a*) 0.12 (*b*) 0.14

(*c*) 0.16 (*d*) 0.18

Electric Field Due to Electric Dipole

159. The electric field due to a short dipole at a distance r, on the axial line, from its mid point is the same as that of electric field at a distance r', on the equatorial line, from its mid-point. Determine the ratio r/r.

160. Two charges of $+25 \times 10^{-9}$ coulomb and -25×10^{-9} coulomb are placed 6 m apart. Find the electric field intensity ratio at points 4 m from the centre of the electric dipole (*i*) on axial line (*ii*) on equatorial line

(*a*) $\frac{1000}{49}$ (*b*) $\frac{49}{1000}$ (*c*) $\frac{500}{49}$ (*d*) $\frac{49}{500}$

161. Electric field due to dipole on axial and equatorial line is perpendicular.

(*a*) True (*b*) False

162. Find angle from the dipole moment where electric field is perpendicular to the dipole moment.

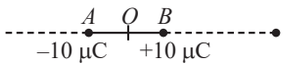
163. Electric field on the axis of dipole must be parallel to electric dipole moment

(*a*) True (*b*) False

164. Electric field on the equatorial line must be opposite to the electric dipole moment

(*a*) True (*b*) False

165. Two charges $\pm 10\ \mu C$ are placed 5.0 mm apart. The electric field at a point P on the axis of the dipole 15 cm away from its centre O on the side of the positive charge, as shown in the figure is:



(a) 2.7×10^5 NC^{-1} (b) 4.13×10^6 NC^{-1}
(c) 3.86×10^6 NC^{-1} (d) 1.33×10^5 NC^{-1}

166. Two ideal dipoles A and B, having their dipole moment P_1 and P_2 respectively are placed on a plane with their centres at O as shown in figure. At point C on axis of dipole A, resultant electric field is making angle of 37° with axis. Ratio of dipole moment of A and B, $\frac{P_1}{P_2}$ is:



(a) $\frac{3}{8}$ (b) $\frac{3}{2}$ (c) $\frac{2}{3}$ (d) $\frac{4}{3}$

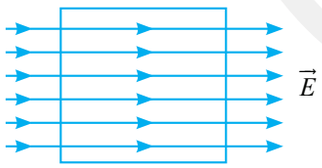
Force Between Two Dipoles

167. The force of interaction between two co-axial short electric dipoles whose centres are R distance apart varies as:
(a) $1/R$ (b) $1/R^2$ (c) $1/R^3$ (d) $1/R^4$

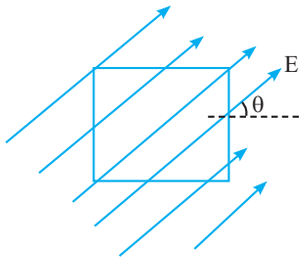
168. The electric force on a point charge situated on the axis of a short dipole is F. If the charge is shifted along the axis to double the distance, the electric force acting will be
(a) $4F$ (b) $\frac{F}{2}$ (c) $\frac{F}{4}$ (d) $\frac{F}{8}$

Force Between Point Charge and Electric Dipole

169. A square surface of side L metre is in the plane of the paper. A uniform electric field $\vec{E}$ (volt/m), also in the plane of the paper, is limited only to the lower half of the square surface (see figure). The electric flux in SI units associated with the surface is



170. A square surface of side L meter in the plane of the paper is placed in a uniform electric field E (volt/m) acting along the same plane at an angle θ with the horizontal side of the square as shown in figure. The electric flux linked to the surface, in units of volt-m, is



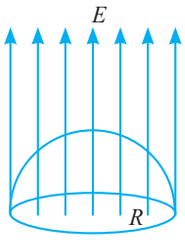
(a) Zero (b) EL^2 (c) $EL^2 \cos\theta$ (d) $EL^2 \sin\theta$

171. If an electric field is given by $10\hat{i} + 3\hat{j} + 4\hat{k}$, calculate the electric flux through a surface of area 10 units lying in yz-plane
(a) 100 units (b) 10 units (c) 30 units (d) 40 units

172. Find incoming and outgoing flux through cylinder.



173. A hemispherical surface of radius R is kept in a uniform electric field E as shown in figure. The flux through the curved surface is



(a) $E2\pi R^2$ (b) $E\pi R^2$ (c) $E4\pi R^2$ (d) Zero

174. A cylinder of radius R and length L is placed in a uniform electric field E perpendicular to the cylinder axis. The outward flux over the surface of the cylinder is given by
(a) $2\pi R^2$E (b) $R^2E/2$ (c) $2\,RLE$ (d) πR^2E

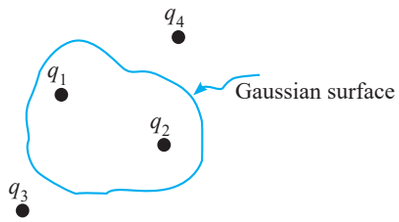
175. A rectangular surface of sides 10 cm and 15 cm is placed inside a uniform electric field of 25 V/m, such that the surface makes an angle of 30° with the direction of electric field. Find the flux of the electric field through the rectangular surface
(a) 0.1675 N/m² C (b) 0.1875 Nm²/C
(c) Zero (d) 0.1075 Nm²/C

176. There is uniform electric field of $8\times10^3\hat{i}$ N/C. What is the net flux (in SI units) of the uniform electric field through a cube of side 0.3 m oriented so that its faces are parallel to the coordinate plane?
(a) $2 \times 8 \times 10^3$ (b) $0.3 \times 8 \times 10^3$
(c) Zero (d) $8 \times 10^6 \times 6$

Gauss Law

177. **Assertion (A):** Four point charges q_1, q_2, q_3 and q_4 are placed as shown in figure. Flux over the Gaussian surface depends only on q_1 and q_2.

Reason (R): Electric field at all points on given Gaussian surface depends on q_1 and q_2 and hence flux also depends only on q_1 and q_2.



(*a*) Assertion (A) is true but Reason (R) is false

(*b*) Assertion (A) is false but Reason (R) is true

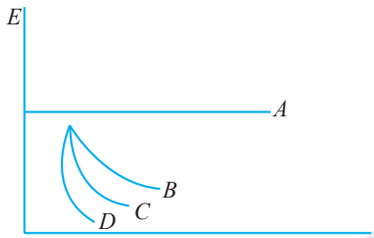
(*c*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A)

(*d*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

178. Gauss law is always applicable to find electric field.

(*a*) True (*b*) False

179. The graph shown illustrates the variation of the electric field (E) as a function of distance (r) for different types of charge distributions. Four curves are labelled as A, B, C, and D, which correspond to different sources of the electric field. Identify the source of electric field.



180. If there were only one type of charge in the universe, then:

(a) $\oint_s E.dS \neq 0$ on any surface

(b) $\oint_s E.dS = 0$ if the charge is outside the surface!

(c) $\oint_s E.dS$ could not be defined.

(d) $\oint_s E.dS = \frac{q}{\varepsilon_0}$ if charges of magnitude q were inside the surface.

Choose the correct statement(s).

(*a*) (a) and (d) (*b*) (a) and (c)

(*c*) (b) and (d) (*d*) (c) and (d)

181. If flux passing through any closed surface is zero then electric field must be zero inside the surface

(*a*) True (*b*) False

182. If charge enclosed by the surface is zero then electric field inside surface must be zero

(*a*) True (*b*) False

183. If charge enclosed by the surface is zero then flux passing through surface must be zero

(*a*) True (*b*) False

184. If $\oint_s \vec{E}.\overrightarrow{dS} = 0$ over a surface, then:

(*a*) The electric field inside the surface is necessarily uniform

(*b*) The number of flux lines entering the surface must be equal to the number of flux lines leaving it

(*c*) The magnitude of electric field on the surface is constant

(*d*) All the charges must necessarily be inside the surface

185. Is Gauss law is valid for both Gaussian surface?



(*a*) Yes (*b*) No

(*c*) can't say

186. Angle between electric field and gaussian surface must be_____ to apply gauss law to find electric field.

187. Gaussian surface can pass through ———— charge distribution and can not pass through ———— charge distribution.

188. Charge is at corner of square plate then find flux passing through square plate.

189. Match the column I with column II.

Column-I		Column-II	
i.	Flux through single surface of cube if q charge is placed at centre of cube	A.	$\frac{q}{6\varepsilon_0}$
ii.	Flux through cube if charge q is placed of centre of one face of cube	B.	$\frac{q}{4\varepsilon_0}$
iii.	Flux through cube if charge q is placed at centre of one edge of cube	C.	$\frac{q}{24\varepsilon_0}$
iv.	Flux through sides of cube that are not in contact with charge q which is placed at one vertex of cube.	D.	$\frac{q}{2\varepsilon_0}$

Choose the correct answer from the options given below:

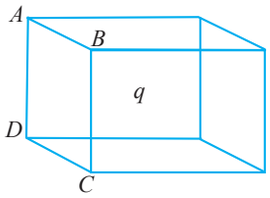
(*a*) (i-A), (ii-D), (iii-B), (iv-C)

(*b*) (i-A), (ii-B), (iii-C), (iv-D)

(*c*) (i-B), (ii-C), (iii-D), (iv-A)

(*d*) (i-B), (ii-D), (iii-A), (iv-C)

190. Charge is inside cube but not at the centre, flux passing through $ABCD$



(*a*) $\frac{q}{\varepsilon_0}$ (*b*) $\phi > \frac{q}{6\varepsilon_0}$

(*c*) $\phi < \frac{q}{6\varepsilon_0}$ (*d*) Both (*b*) and (*c*)

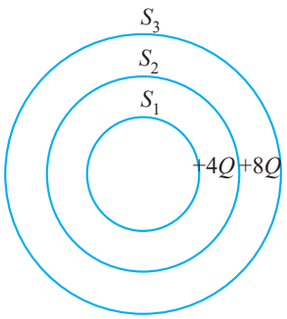
191. A point charge is placed at the centre of spherical Gaussian surface. Electric flux ϕ_E changed if

(*a*) The sphere is replaced by a cube of same or different volume?

(*b*) The charge is oved off-centre in the original sphere, still remaining inside?

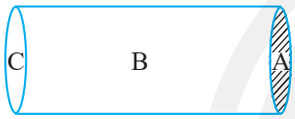
(*c*) The charge is moved just outside the original sphere?
(*d*) A second charge is placed near the outside of the original sphere?
(*e*) A second charge is placed inside the Gaussian surface?

192. S_1, S_2, S_3 are three concentric spherical shells, as shown in figure. S_1 and S_2 have charge $4Q$ and $8Q$ respectively. whereas S_3 is uncharged. Find the ratio of the flux passing through S_1 to that through S_3



(*a*) 1 : 3 (*b*) 1 : 2 (*c*) 2 : 1 (*d*) 1 : 6

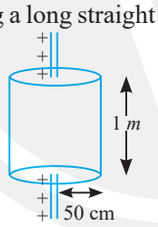
193. A hollow cylinder has a charge q coulomb within it. If ϕ is the electric flux in units of voltmetre associated with the curved surface B, the flux linked with the plane surface A in unit of voltmetre will be (charge is symmetrically placed within it)



(*a*) $\frac{q}{\varepsilon_0}-\phi$ (*b*) $\frac{1}{2}\left(\frac{q}{\varepsilon_0}-\phi\right)$

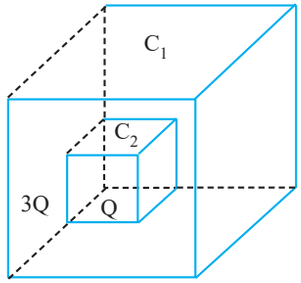
(*c*) $\frac{q}{2\varepsilon_0}$ (*d*) $\frac{\phi}{3}$

194. Electric charge is uniformly distributed along a long straight wire of radius 1 mm. The charge per cm length of the wire is Q coulomb. Another cylindrical surface of radius 50 cm and length 1 m symmetrically encloses the wire as shown in the figure. The total electric flux passing through the cylindrical surface is:



(*a*) $\frac{Q}{\varepsilon_0}$ (*b*) $\frac{100Q}{\varepsilon_0}$ (*c*) $\frac{10Q}{\pi\varepsilon_0}$ (*d*) $\frac{100Q}{\pi\varepsilon_0}$

195. C_1 and C_2 are two hollow concentric cubes enclosing charges 2Q and 3Q respectively as shown in figure. The ratio of electric flux passing through C_1 and C_2 is:

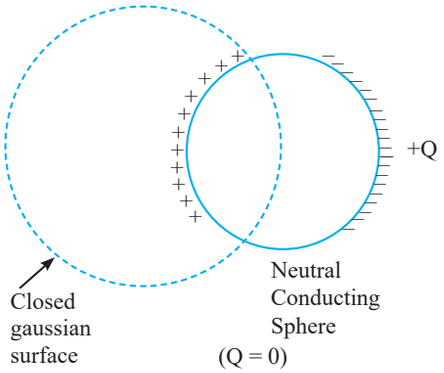


(*a*) 3 : 2 (*b*) 5 : 2 (*c*) 2 : 5 (*d*) 2 : 3

196. Two charges of $5Q$ and $-2Q$ are situated at the points $(3a, 0)$ and $(-5a, 0)$ respectively. The electric flux through a sphere of radius '$4a$' having center at origin is:

(*a*) $2Q/\varepsilon_0$ (*b*) $7Q/\varepsilon_0$ (*c*) $3Q/\varepsilon_0$ (*d*) $5Q/\varepsilon_0$

197. Find flux passing through closed gaussian surface, when $+Q$ point charge placed near to conducting neutral sphere



(*a*) Zero (*b*) +ve (*c*) –ve (*d*) can't say

198. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): If an electric dipole of dipole moment 30×10^{-5} Cm is enclosed by a closed surface, the net flux coming out of the surface will be zero.

Reason (R): Electric dipole consists of two equal and opposite charges. In the light of above, statements, choose the correct answer from options given below:

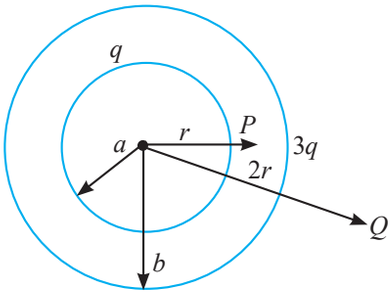
(*a*) Assertion (A) is true but Reason (R) is false
(*b*) Assertion (A) is false but Reason (R) is true
(*c*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A)
(*d*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

199. Choose incorrect statement

(1) Electric line of force entering into a gaussian surface provide negative flux.
(2) A charge 'q' is placed at centre of cube. The flux through all faces will be same.
(3) In uniform electric field net flux through closed gaussian surface containing no net charge is zero.
(4) When electric field is parallel to gaussian surface, it provides a finite non- zero flux

(*a*) (3) and (4) only
(*b*) (2) and (4) only
(*c*) (4) Only
(*d*) (1) and (3) only

200. The electric field intensity at P and Q, in the shown arrangment, are in the ratio



(*a*) 1 : 2 (*b*) 2 : 1 (*c*) 1 : 1 (*d*) Zero

201. A hollow metal sphere of radius R is uniformly charged. The electric field due to the sphere at a distance r from the centre

(*a*) decreases as r increases for $r < R$ and for $r > R$

(*b*) increases as r increases for $r < R$ and for $r > R$

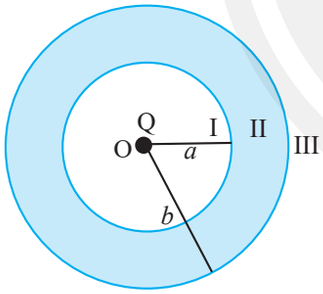
(*c*) zero as r increases for $r < R$, decreases as r increases for $r > R$

(*d*) zero as r increases for $r < R$, increases as r increases for $r > R$

202. The electric field at a distance 3R/2 from the centre of a charged conducting spherical shell of radius R is E. The electric field at a distance R/2 from the centre of the sphere is

(*a*) Zero (*b*) E (*c*) $\frac{E}{2}$ (*d*) $\frac{E}{3}$

203. As shown in the figure, a point charge Q is placed at the centre of conducting spherical shell of inner radius and outer radius b. The electric field due to charge Q in three different regions I, II and III is given by: (I: $r < a$, II: $a < r < b$, III : $r > b$)



(*a*) $E_1 = 0, E_{11} = 0, E_{111} \neq 0$

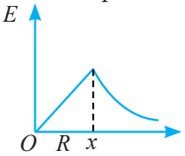
(*b*) $E_1 \neq 0, E_{11} = 0, E_{111} = 0$

(*c*) $E_{10}, E_{11} = 0, E_{111} \neq 0$

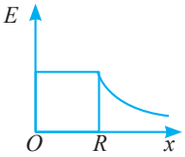
(*d*) $E_1 = 0, E_{11} = 0, E_{111} = 0$

204. An isolated solid metal sphere of radius R is given an electric charge. Which of the graphs below best shows the way in which the electric field E varies with distance x from the centre of the sphere.

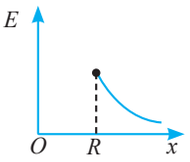
(*a*)



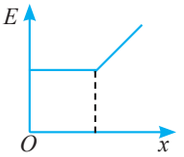
(*b*)



(*c*)



(*d*)



205. A non conducting solid sphere of radius R is uniformly charged. The electric field due to the sphere at a distance r from the centre

(*a*) decreases as r increases for $r < R$ and for $r > R$

(*b*) increases as r increases for $r < R$ and decreases as r increases for $r > R$

(*c*) zero as r increases for $r < R$ decreases as r increases for $r > R$

(*d*) zero as r increases for $r < R$ increases as r increases for $r > R$

206. Consider an atom with atomic number Z as consisting of a positive point charge at the centre and surrounded by a distribution of negative electricity uniformly distributed within a sphere of radius R. The electric field at a point inside the atom at a distance r from the centre is

(*a*) $\frac{Ze}{4\pi\varepsilon_0}\left[\frac{1}{r^2}-\frac{r}{R^3}\right]$ (*b*) $\frac{Ze}{4\pi\varepsilon_0}\left[\frac{1}{r^2}+\frac{r}{R^3}\right]$

(*c*) $\frac{2Ze}{4\pi\varepsilon_0 r^2}$ (*d*) Zero

207. A spherical conductor of radius 10 cm has a charge of 3.2 $\times 10^{-7}$C distributed uniformly. What is the magnitude of electric field at a point 15 cm from the centre of the sphere?

$\left(\frac{1}{4\pi\varepsilon_0}=9\times10^9\ N\,m^2/C^2\right)$

(*a*) $1.28 \times 10^4 \times$ N/C (*b*) $1.28 \times 10^5 \times$ N/C

(*c*) $1.28 \times 10^6 \times$ N/C (*d*) $1.28 \times 10^7 \times$ N/C

208. A solid metallic sphere has a charge $+3Q$. Concentric with this sphere is a conducting spherical shell having charge $-Q$. The radius of the sphere is a and that of the spherical shell is b(b > a). What is the electric field at a distance $R(a<R<b)$ from the centre:

(*a*) $\frac{Q}{2\pi\varepsilon_0 R}$ (*b*) $\frac{3Q}{2\pi\varepsilon_0 R}$ (*c*) $\frac{3Q}{4\pi\varepsilon_0 R}$ (*d*) $\frac{4Q}{4\pi\varepsilon_0 R}$

209. Consider the uniformly charged spherical shell of Radius R carrying charge Q and charge 'q' placed at distance r from the centre of shell. If charge q is experiencing a force F, then identify the correct statement from following.

(*a*) $\frac{1}{4\pi\varepsilon_0}\frac{qQ}{R^2} > f > 0$ for $r < R$

(*b*) $f=\frac{1}{4\pi\varepsilon_0}\frac{qQ}{r^2}$ for $r < R$

(*c*) $f=\frac{1}{4\pi\varepsilon_0}\frac{qQ}{r^2}$ for all r

(*d*) $f=\frac{1}{4\pi\varepsilon_0}\frac{qQ}{r^2}$ for $r < R$

210. **Assertion (A):** If two spherical conductors of different radii have the same surface charge densities, then their electric field intensities will be equal.

Reason (R): Surface charge density = Total charge/Area

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A)

(*c*) Assertion (A) is true but Reason (R) is false

(*d*) Assertion (A) is false and Reason (R) is true

211. A dipole with an electric moment $\vec{p}$ is located at a distance r from a long thread charged uniformly with a linear charge density λ. Find the force F acting on the dipole if the vector $\vec{p}$ is oriented along the thread.

(*a*) $\frac{p\lambda}{2\pi\varepsilon_0 r^2}$ (*b*) $\frac{p\lambda}{2\pi\varepsilon_0 r}$

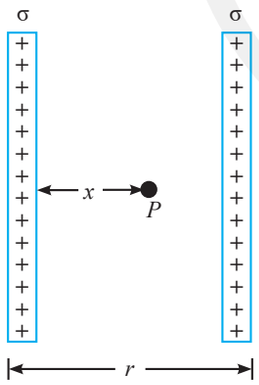
(*c*) $\frac{p}{2\pi\varepsilon_0 r\lambda}$ (*d*) Zero

212. Two parallel infinite line charges with linear charge densities +λC/m and –λC/m are placed at a distance of $2R$ in free space. What is the electric field mid-way between the two line charges?

(*a*) $\frac{\lambda}{2\pi\varepsilon_0 R} N/C$ (*b*) Zero

(*c*) $\frac{2\lambda}{\pi\varepsilon_0 R} N/C$ (*d*) $\frac{\lambda}{\pi\varepsilon_0 R} N/C$

213. For two infinitely long charged parallel sheets, the electric field at P will be



(*a*) $\frac{\sigma}{2x} - \frac{\sigma}{2(r-x)}$ (*b*) $\frac{\sigma}{2\varepsilon_0 x} - \frac{\sigma}{2\pi(r-x)\varepsilon_0}$

(*c*) $\frac{\sigma}{\varepsilon_0}$ (*d*) Zero

214. If a small sphere of mass m and charge q is hung from a silk thread at an angle θ with the surface of a vertical charged conducting plate, then for equilibrium of sphere, the surface charge density of the plate is

(*a*) $\varepsilon_0\left(\frac{mg}{q}\right)\tan\theta$ (*b*) $\varepsilon_0\left(\frac{2mg}{q}\right)\tan\theta$

(*c*) $\varepsilon_0(mgq)\tan\theta$ (*d*) $\varepsilon_0\left(\frac{mg}{3q}\right)\tan\theta$

215. Same charge Q given on both the plate conducting and Nonconducting of Area 'A' then find Ratio of electric field very near to both the plates.

216. Ratio of electric field due to conducting and non-conducting sheet having equal charge density is 2 : 1. [True/False]

(*a*) True (*b*) False

217. An electron is rotating around an infinite positive linear charge in a circle of radius 0.1 m, if the linear charge density is 1 μC/m, then the velocity of electron in m/s will be

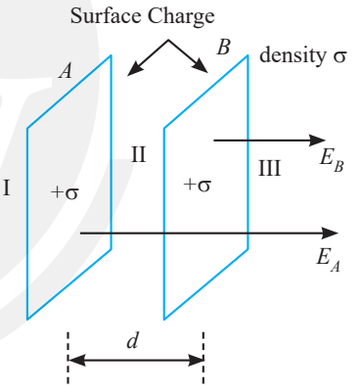
(*a*) 0.562×10^7 (*b*) 5.62×10^7

(*c*) 562×10^7 (*d*) 0.0562×10^7

218. Two long thin charged rods with charge density λ each are placed parallel to each other at a distance d apart. The force per unit length exerted on one 1 rod by the other will be

(*a*) $\frac{k2\lambda}{d}$ (*b*) $\frac{k2\lambda^2}{d}$ (*c*) $\frac{k2\lambda}{d^2}$ (*d*) $\frac{k2\lambda^2}{3d^2}$

219. Let s be the uniform surface charge density of two infinite thin plane sheets shown in figure. Then the electric field in three different region E_1, E_{11} and E_{111} are:



(*a*) $\vec{E}_1 = 0, \vec{E}_{11} = \frac{\sigma}{\varepsilon_0}\hat{n}, E_{111} = 0,$

(*b*) $\vec{E}_1 = \frac{\sigma}{2\varepsilon_0}\hat{n}, \vec{E}_{11} = 0, E_{111} = \frac{\sigma}{2\varepsilon_0}\hat{n}$

(*c*) $\vec{E}_1 = \frac{-\sigma}{\varepsilon_0}\hat{n}, \vec{E}_{11} = 0, E_{111} = \frac{\sigma}{\varepsilon_0}\hat{n}$

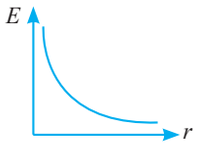
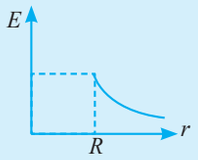
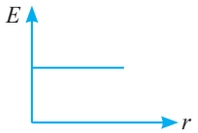
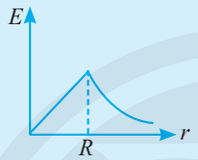
(*d*) $\vec{E}_1 = \frac{2\sigma}{\varepsilon_0}\hat{n}, \vec{E}_{11} = 0, E_{111} = \frac{2\sigma}{\varepsilon_0}\hat{n}$

220. Two parallel metal plates having charges $+Q$ and $-Q$ faces each other at a certain distance between them. If the plates are now dipped in kerosene oil tank, the electric field between the plates will:

(*a*) become zero (*b*) increase

(*c*) decrease (*d*) remain same

221. Match the Column-I with Column-II.

Column-I		Column-II	
(i)	Electric field due to solid conducting sphere	A.	
(ii)	Electric field due to infinite line charge.	B.	
(iii)	Electric field due to solid non-conducting sphere.	C.	
(iv)	Electric field due to infinite thin plate.	D.	

Choose the correct answer from the options given below:

(*a*) (i-B), (ii-A), (iii-D), (iv-C)

(*b*) (i-A), (ii-B), (iii-C), (iv-D)

(*c*) (i-B), (ii-C), (iii-D), (iv-A)

(*d*) (i-B), (ii-D), (iii-A), (iv-C)

222. An electric dipole consists of two opposite charges each of magnitude 1μC separated by a distance of 2 cm. The dipole is placed in an external field of 10^5 N/C. The maximum torque on the dipole is

(*a*) 2×10^{-4} Nm (*b*) 2×10^{-3} Nm

(*c*) 4×10^{-3} Nm (*d*) 10^{-3} Nm

223. The torque τ acting on an electric dipole of dipole moment in an electric field $\vec{E}$ is

(*a*) $\vec{\tau} = \vec{p} \cdot \vec{E}$ (*b*) $\vec{\tau} = \vec{p} \times \vec{E}$ (*c*) $\vec{\tau} = p\vec{E}$ (*d*) $\vec{\tau} = \vec{p}E$

224. An electric dipole is placed at an angle of 30° with an electric field of intensity 10^7 NC^{-1}. It experiences a torque equal to 15×10^2 N-m. What will be magnitude of charge on dipole if it is 5cm long?

(*a*) 1 mC (*b*) 2 mC (*c*) 6 mC (*d*) None

225. The frequency of oscillation of an electric dipole moment having dipole moment *p* and rotational inertia *I*, oscillating in a uniform electric field *E* is given

(*a*) $(1/2\pi)\sqrt{I/pE}$ (*b*) $(1/2\pi)\sqrt{pE/I}$

(*c*) $(2\pi)\sqrt{pE/I}$ (*d*) $(2\pi)\sqrt{I/pE}$

226. An electric dipole is placed at an angle of 30° with an electric field intensity 2×10^5 N/C. It experiences a torque equal to 4 N m. The charge on the dipole, if the dipole length is 2 cm, is

(*a*) 8 *mC* (*b*) 2 *mC*

(*c*) 5 *mC* (*d*) 7 *mC*

Potential Energy Stored in the Dipole

227. An electric dipole when placed in a uniform electric field E will have minimum potential energy, when the angle made by dipole moment with field E is

(*a*) π (*b*) $\frac{3\pi}{2}$ (*c*) Zero (*d*) $\frac{\pi}{2}$

228. Dipole can be in equilibrium in two orientation in electric field one is stable and other is unstable

(*a*) True (*b*) False

229. Path of charge particle in electric field may be circle

(*a*) True (*b*) False

Work Done in Rotating a Dipole in Uniform Electric Field

230. An electric dipole moment is lying along a uniform electric field E. The work done in rotating the dipole by 90° is

(*a*) $\sqrt{2}pE$ (*b*) $\frac{pE}{2}$

(*c*) $2pE$ (*d*) pE

231. An electric dipole of moment p is lying along a uniform electric field *E*. The work done in rotating the dipole by 180° is

(*a*) pE (*b*) $+2\,pE$

(*c*) $+2\,pE$ (*d*) $\frac{\phi E}{\sqrt{2}}$

232. An electric dipole placed in a uniform electric field experiences maximum moment of couple when the dipole is placed

(*a*) Against the direction of the field

(*b*) Towards the electric field

(*c*) Perpendicular to the direction of the field

(*d*) At 135° to the direction of the field

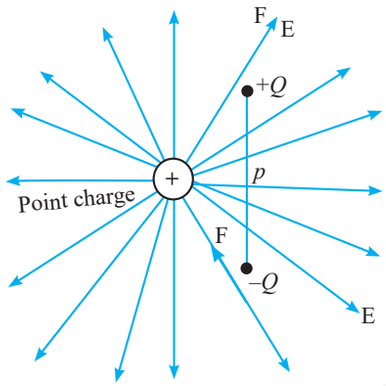
233. Work done by electric field (*E*) to rotate the dipole of dipole moment ϕ from Stable to unstable equilibrium ?

234. Work done to rotate the dipole from unstable equilibrium to 60° is √3 J then, find external torque required to keep the dipole at this position ?

Dipole Placed in Non-Uniform Electric Field

235. Dipole is placed in non-uniform electric filed due to point charge as shown in figure

Direction of net force

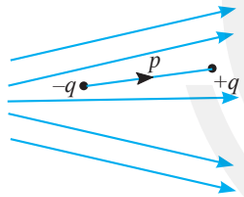


(*a*) along +*x* axis (*b*) along –*x* axis

(*c*) along +*y* axis (*d*) along +*y* axis

(*e*) along +*y* axis

236. The figure shows electric field lines in which an electric dipole *p* is placed as shown. Which of the following statements is correct?



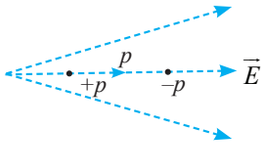
(*a*) The dipole will not experience any force

(*b*) The dipole will experience a force towards the right

(*c*) The dipole will experience a force towards the the left

(*d*) The dipole will experience a force upwards

237. A dipole is placed in an electric field as shown. In which direction will it move?



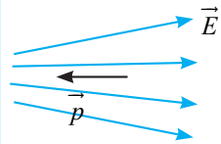
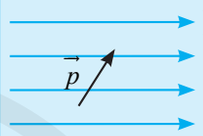
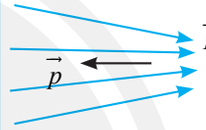
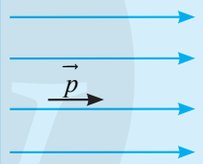
(*a*) Towards the left as its potential energy will decrease

(*b*) Towards the right as its potential energy will increase

(*c*) Towards the left as its potential energy will increase

(*d*) Towards the right as its potential energy will decrease

238. Match the column-I with column-II.

Column-I		Column-II	
I.		A.	Dipole is in Translational as well as in rotational equilibrium
II.		B.	Dipole is not in equilibrium and it moves towards left side.
III.		C.	Dipole is in translational equilibrium but not in rotational equilibrium.
IV.		D.	Dipole is not in equilibrium and it moves towards right side.

Polar & Non-Polar Molecules

239. Polar molecules are the molecules

(*a*) That acquires a dipole moment only when the magnetic field is absent

(*b*) Has a permanent electric dipole moment

(*c*) Has zero dipole moment

(*d*) That acquire a dipole moment only in the presence of an electric field due to displacement of chargesa

MR* CORNER

1. Match the facts given in Column-I with the Column-II:

Column-I		Column-II	
I.	Charge cannot exist	a.	Without charge
II.	Mass can exist	b.	Without mass
III.	Charge is	c.	Not conserved
IV.	Mass is	d.	Conserved

(*a*) I-b, II-a, III-d, IV-c (*b*) I-b, II-a, III-c, IV-d

(*c*) I-b, II-d, III-a, IV-c (*d*) I-b, II-c, III-d, IV-a

2. A uniformly charged ring is placed in x - z plane as shown in figure. A point charge ($+q$ is placed on y-axis and it is in equilibrium. Choose the correct statement.

(*a*) If $y = \frac{R}{\sqrt{2}}$, nature of equilibrium is stable

(*b*) If $y > \frac{R}{\sqrt{2}}$, nature of equilibrium is unstable

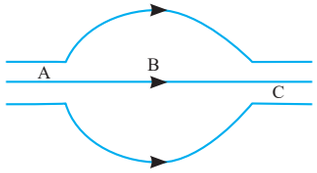
(*c*) If $y < \frac{R}{\sqrt{2}}$, nature of equilibrium is unstable

(*d*) Is $y = \frac{R}{\sqrt{2}}$, is the position where electric force experienced by $+q$ is minimum

3. Two equal point charges A and B are R distance apart. A third point charge placed on the perpendicular bisector at a distance 'x' from the centre. They experience maximum electrostatic force when:

(*a*) $x = \frac{R}{2\sqrt{2}}$ (*b*) $x = \frac{R}{\sqrt{2}}$ (*c*) $x = R\sqrt{2}$ (*d*) $x = 2\sqrt{2}R$

4. Given figure shows some of the electric field lines corresponding to an electric field. The figure suggests that electric field strength at A point is____as compared to point C but it is_____as compared to point B



5. **Statement I:** The net force on the dipole due to electric field generated by point change may be Zero.

Statement II: Net torque on dipole in a electric field generated by infinite large rod must be zero when dipole is placed perpendicularly to rod.

6. The _____ acting on A proton and an electron placed in uniform electric field will be repel.

(*a*) Acceleration

(*b*) Magnitude of their acceleration

(*c*) Magnitude of the force

(*d*) Electric forces

7. The _____ of the universe is constant.

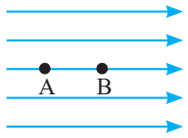
(*a*) Total number of charged particles.

(*b*) Total negative charge.

(*c*) Total charge

(*d*) Total positive

8. A point charge is brought in an uniform electric field at point A. The electric field at nearly print B will_____.



(*a*) Must increase

(*b*) Must decrease

(*c*) Increase if point charge is negative.

(*d*) Increase if point charge is positive.

9. If a _____ charge is shifted from a low-potential region to a high-potential region, the electric potential energy increases.

(*a*) Negative

(*b*) May be Positive of Negative

(*c*) Positive

(*d*) Electric Potential Energy cannot increase whatever charge May be

10. **Assertion (A):** If a spherical cavity is made inside a uniformly charged sphere, then electric field inside the cavity is uniform.

Reason (R): The electric field inside the cavity is independent of radius of cavity, but depends on radius of sphere.

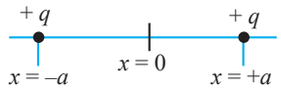
(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

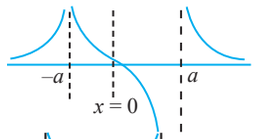
(*b*) Both Assertion (A) and Reason (R) are true and Reason (R) is NOT the correct explanation of Assertion (A).

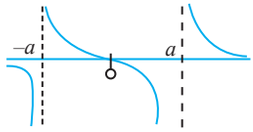
(*c*) Assertion (A) is true but Reason (R) is false

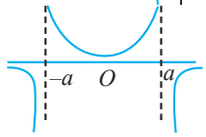
(*d*) Assertion (A) is false but Reason (R) is true.

11. Consider the charge configuration shown in the figure. Variation of electric field along x-axis is



(*a*) 

(*b*) 

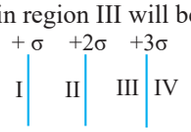
(*c*) 

(*d*) None of these

12. **Statement I:** A charge Q is uniformly distributed over a large plastic plate. The electric field at a point P close to the centre of plate is 10 Vm^{-1}. If the plastic plate is replaced by copper plate of same geometrical dimensions & carrying same charge Q. The electric field at point P will remain same.

Statement II: A Metallic particle having no net change is placed near a finite metal plate carrying a positive charge. The electric force on particle will be zero

13. Three large non-conducting plane sheets of charge with charge densities σ, 2σ and 3σ are arranged as shown. The net electric field in region III will be

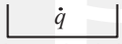


(*a*) $\frac{3\sigma}{\varepsilon_0}$ (towards left) (*b*) $\frac{2\sigma}{\varepsilon_0}$ (towards left)

(*c*) 0 (*d*) $\frac{3\sigma}{\varepsilon_0}$ (towards right)

14. **Statement I:** Electric charges are distributed in small volume. The flux of the electric field through a spherical surface of radius 10 cm. Surrounding the total charge is 25 Vm. The flux over a concentric sphere of radius 20cm will be 50 Vm.

Statement II: A charge *q* is placed at the centre of the open end of a cylindrical vessel as shown in figure. The flux of electrical field through surface of the vessel will be zero as gauss law is only valid for closed surfaces.



15. **Statement-I:** Coulombic force follows the principle of superposition.

Statement-II: Coulombic force is directly proportional to the charges on both particle

(*a*) Statement-I is true, statement-II is true and statement-II is correct explanation for statement-I.

(*b*) Statement-I is true, statement-II is true and statement-II is NOT the correct explanation for statement-I

(*c*) Statement-I is true, statement-II is false.

(*d*) Statement-1 is false, statement-II is true.

16. Two identical charged spheres suspended from a common point by two mass-less strings of length *l* are initially at a distance *d* ($d << l$) apart because of their mutual repulsion. The charge begins to leak from both the spheres at a constant rate. As a result the charge approach each other with a velocity *v*. Then as a function of distance *x* between them.

(*a*) $v \propto x^{-1}$ (*b*) $v \propto x^{\frac{1}{2}}$ (*c*) $v \propto x$ (*d*) $v \propto x^{-\frac{1}{2}}$

17. **Statement-I:** In a given situation of arrangement of charges, an extra charge is placed outside the Gaussian surface. In the Gauss Theorem

$$\oint \vec{E}.d\vec{s} = \frac{Q_{in}}{\epsilon_0}$$

Q_{in} remains unchanged where $\vec{E}$ as electric field at the site of the element is changed.

Statement-II: Electric field E at any point on the Gaussian surface is due to inside charge only.

(*a*) **Statement-I** is true, statement-II is true and statement-II is correct explanation for statement-1

(*b*) **Statement-I** is true, statement-II is true and statement-II is NOT the correct explanation for statement-I

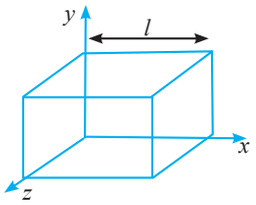
(*c*) **Statement-I** is true, statement-II is false

(*d*) **Statement-I** is false, statement-II is true.

18. **Statement I:** The flux of electric field through closed surface due to all charges is equal to flux due to charges enclosed by surface.

Statement II: The electric field calculated by gauss's law, is field due to the charges inside the gaussian surface.

19. Electric field of a region varies as $\vec{E} = bx\,\hat{i}\ NC^{-1}$. Find the net charge enclosed by the cube as shown in figure



(*a*) $\frac{bl^3}{2\varepsilon_0}$ (*b*) $\frac{bl^3}{3\varepsilon_0}$

(*c*) $bl^3\varepsilon_0$ (*d*) $\frac{bl^3\varepsilon_0}{2}$

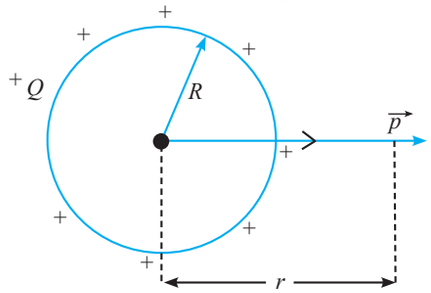
20. **Statement I:** If a positive point charge Q is brought near an isolated metal cube, then the interior of cube becomes positively charged and the surface becomes negatively charged.

Statement II: If A positive point charge Q is brought near an isolated metal cube, then interior remains charge free and surface gets non uniform charge distribution.

21. **Statement I:** If flux of electric field through a closed surface is zero, the electric field may be zero everywhere on the surface.

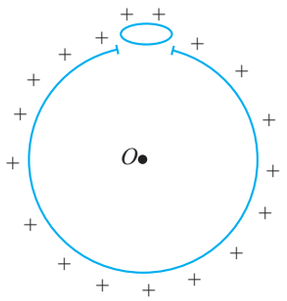
Statement II: If flux of electric field through a closed surface is zero, then the change in the vicinity of surface must be zero.

22. A short electric dipole is placed along a radial line of a uniformly charged sphere as shown. If the force experienced by dipole at $r = 2R$ and $r = 4R$ are F_1 and F_2 respectively, then F_2/F_1 is



(*a*) 1/8 (*b*) 1/4 (*c*) 1/16 (*d*) 1/32

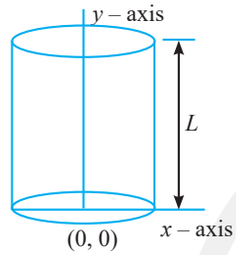
23. A conducting charged spherical shell has a tiny hole cut into its surface. If surface charge density near the hole is σ then electric field in the hole is:-



(a) $\frac{\sigma}{\epsilon_0}$, directed inwards (b) $\frac{\sigma}{\epsilon_0}$, directed outwards

(c) $\frac{\sigma}{2\epsilon_0}$, directed inwards (d) $\frac{\sigma}{2\epsilon_0}$, directed outwards

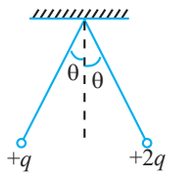
24. A long charged wire is kept on $+y$-axis, with one end at origin. Its linear charge density is given as $\lambda = \lambda_0 y$. Calculate electric flux passing through the cylinder.

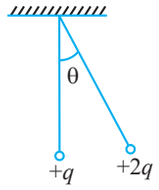


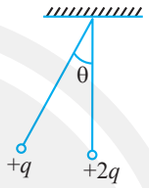
(a) $\frac{\lambda_0 L_0}{\epsilon_0}$ (b) $\frac{\lambda_0 L^2}{2\epsilon_0}$

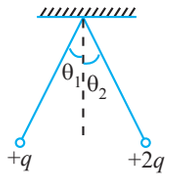
(c) $\frac{2\lambda_0 L^2}{\epsilon_0}$ (d) None

25. Two metal spheres of same mass are suspended from a common point by a light insulating string. The length of each string is same. The spheres are given electric charges $+q$ on one and $+2q$ on the other. Which of the following diagram best shows the resulting positions of spheres?

(a)


(b)


(c)


(d)


Answer Key

1. (a) **2.** (a) **3.** (c) **4.** (a) **5.** (c) **6.** (d) **7.** (b) **8.** (b) **9.** (b) **10.** (b)

11. $S_{neutron} = 0$, $S_{proton} = 2\ (S_{\alpha\ particle})$ **12.** (c) **13.** (a, b, d) **14.** (a, d) **15.** (c) **16.** $Q = 16 \times 10^{-19}$ C **17.** (c)

18. $n = 6.25 \times 10^{18}$ **19.** (a) **20.** 12 **21.** (d) **22.** 56.8×10^{-25} kg **23.** 6.25×10^8 seconds

24. 5.49×10^{30}C **25.** (d) **26.** (b) **27.** (d) **28.** (c) **29.** (d) **30.** (c) **31.** (c) **32.** (d)

33. 8C **34.** (b) **35.** (a) **36.** (d) **37.** (b) **38.** (a) **39.** (a) **40.** (c) **41.** (b) **42.** (c)

43. (a) **44.** (a) **45.** (c) **46.** 2.3×10^{-28} N **47.** $\simeq 10^{36}$ **48.** 16 Newton **49.** 10^{-5} Newton **50.** (d)

51. (a) **52.** (a) **53.** (c) **54.** 16 : 9 **55.** (d) **56.** (b) **57.** (c) **58.** (a) **59.** $v = \sqrt{\frac{q^2}{16\pi\varepsilon_0 mr}}$

60. (c) **61.** (b) **62.** $F = kq\lambda\left[\frac{1}{a} - \frac{1}{a+L}\right]$ **63.** (a) **64.** (a) **65.** (c) **66.** (d) **67.** 0

68. $\frac{20kqQ}{l^2}$ **69.** $\frac{4\sqrt{2}kQq}{l^2}$ **70.** (b) **71.** $\frac{kQ^2}{\ell^2}\left(\frac{2\sqrt{2}+1}{2}\right)$ **72.** (b) **73.** (c)

74. $q = \frac{-Q}{2\sqrt{2}}$ **75.** $\frac{-q\left(2\sqrt{2}+1\right)}{4}$ **76.** $\frac{4kq_0q}{3}$ **77.** $\frac{4}{5}kqq_0$ **78.** (a) **79.** $\frac{2kQqr}{(a^2+r^2)^{3/2}}$

80. (a) **81.** $\frac{9\times10^9\times10[3\hat{i}+4\hat{j}+5\hat{k}]}{(5\sqrt{2})^3}$ **82.** $\frac{108\times10^9(3\hat{i}+4\hat{j})}{125}$ **83.** (b) **84.** (b) **85.** (a) **86.** (a)

87. (i) 2m from Q, (ii) 7.5m form 3Q, (iii) $\frac{80}{9}m$ from 16 Q, (iv) $\frac{10}{\sqrt{2}+1}$ from –4Q **88.** 15 m from 25 Q

89. (i) $\frac{10}{3}m$ from -*Q*, (ii) 20 m form +8Q **90.** (*b*) **91.** Unstable equilibrium along *x*–axis

92. Stable equilibrium along *y* – axis **93.** (*c*) **94.** (*c*) **95.** (*a*) **96.** (*b*) **97.** (*a*) **98.** (*b*) **99.** (*b*)

100. (*d*) **101.** (*d*) **102.** (*i*) 0, (*ii*) $\frac{\sqrt{2}KQ}{r^2}$, (*iii*) $\frac{2KQ}{r^2}$ **103.** 5m from + Q **104.** (*b*) **105.** (*a*) **106.** (*d*)

107. Zero **108.** $\frac{4\sqrt{2}kQ}{\ell^2}$ **109.** 0 **110.** $\frac{Q}{\pi\varepsilon_0\ell^2}$ **111.** (*c*) **112.** $\frac{2\sqrt{2}kq}{\ell^2}$ **113.** $\frac{KQ}{a^2}\frac{\left(2\sqrt{2}+1\right)}{2}$ **114.** (*c*)

115. $4kq/3a^2$ **116.** $R/\sqrt{2}$ **117.** $x=\frac{R}{2\sqrt{2}}$ **118.** (*c*) **119.** (*c*) **120.** (*d*) **121.** (*a*) **122.** (*a*) **123.** (*a*)

124. $\frac{-5}{2}$ **125.** D **126.** (*d*) **127.** (*a*) **128.** (*d*) **129.** (*c*) **130.** (*a*) **131.** (*d*) **132.** (*a*) **133.** (*b*)

134. (*b*) **135.** (*a*) **136.** 1:1 **137.** (*b*) **138.** 1:1 **139.** (*a*) **140.** (*d*) **141.** (*b*) **142.** (*a*) **143.** (*b*)

144. (*c*) **145.** (*b*) **146.** (*c*) **147.** (*b*) **148.** (*i*) 2 : 1 : 1, (*ii*)1 : 1 : 2, (*iii*) 1 : 2 : 8 **149.** (*c*)

150. (*d*) **151.** (*b*) **152.** (*d*) **153.** (*i*) $\sqrt{3}qa$, (*i*) P = $\sqrt{2}ql$, (*iii*) $\sqrt{5}qa$ **154.** $\sqrt{3}ql$ **155.** (*d*) **156.** (*a*)

157. (*c*) **158.** (*a*) **159.** $2^{\frac{1}{3}}:1$ **160.** (*a*) **161.** (*b*) **162.** $\theta=\tan^{-1}(\sqrt{2})$ **163.** (*b*) **164.** (*a*)

165. (*a*) **166.** (*c*) **167.** (*d*) **168.** (*d*) **169.** Zero **170.** (*a*) **171.** (*a*) **172.** $\phi_{in}=-E\pi R^2$, $\phi_{out}=E\pi R^2$

173. (*b*) **174.** (*c*) **175.** (*b*) **176.** (*c*) **177.** (*a*) **178.** (*b*)

179. A = Due to infinite Sheet, B = Due to infinite Rod, C = Charged Particle (Point Charge), D = Dipole **180.** (*c*) **181.** (*b*)

182. (*b*) **183.** (*a*) **184.** (*b*) **185.** (*a*) **186.** [0°, 180° or 90°] **187.** [Continuous, Discretel] **188.** Zero

189. (*a*) **190.** (*c*) **191.** (*c,e*) **192.** (*a*) **193.** (*b*) **194.** (*b*) **195.** (*c*) **196.** (*d*) **197.** (*b*) **198.** (*d*)

199. (*c*) **200.** (*b*) **201.** (*c*) **202.** (*a*) **203.** (*c*) **204.** (*c*) **205.** (*b*) **206.** (*a*) **207.** (*b*) **208.** (*c*)

209. (*b*) **210.** (*b*) **211.** (*d*) **212.** (*d*) **213.** (*d*) **214.** (*a*) **215.** 1:1 **216.** (*a*) **217.** (*b*) **218.** (*b*)

219. (*c*) **220.** (*b*) **221.** (*a*) **222.** (*a*) **223.** (*b*) **224.** (*c*) **225.** (*b*) **226.** (*b*) **227.** (*c*) **228.** (*a*)

229. (*b*) **230.** (*d*) **231.** (*d*) **232.** (*c*) **233.** 2pE **234.** τ = –1 N.m **235.** (*c*) **236.** (*c*) **237.** (*d*)

238. [1-D, 2-C, 3-B, 4-A] **239.** (*b*)

MR* CORNER

1. (*a*) **2.** (*c*) **3.** (*a*) **4.** Equal, Greater **5.** I. - False, II. - True **6.** (*c*) **7.** (*c*) **8.** (*d*)

9. (*c*) **10.** (*c*) **11.** (*b*) **12.** I. - True, II. - False **13.** (*c*) **14.** I. - False, II. - False **15.** (*d*) **16.** (*d*)

17. (*c*) **18.** I. True, I. False **19.** (*c*) **20.** I. - False, II. - True **21.** I- True, II. - False **22.** (*a*) **23.** (*d*)

24. (*b*) **25.** (*a*)